Supplementary Data

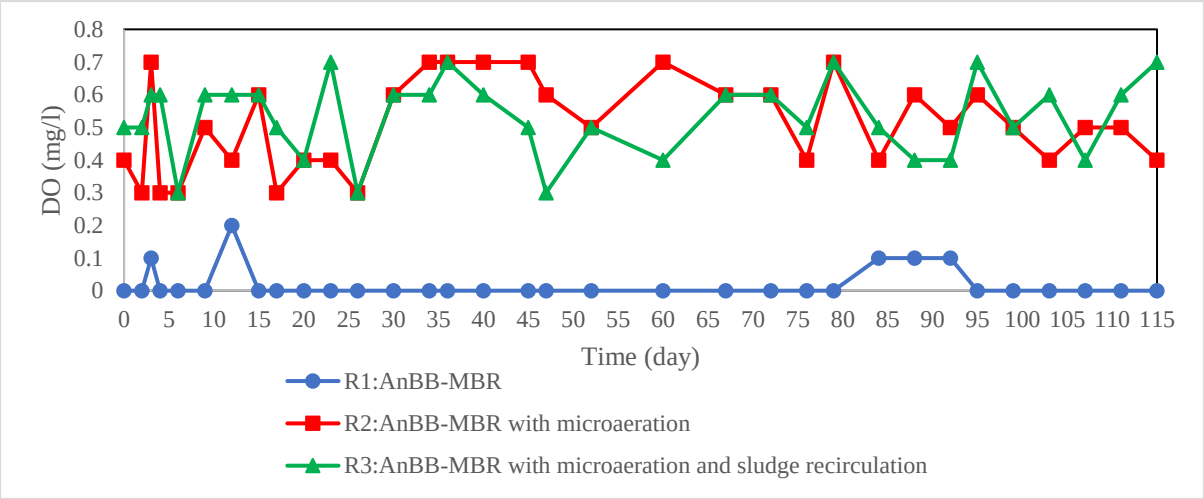


Fig. S1 Dissolved oxygen (DO) in C3 compartment of the three AnBB-MBR systems during the operation phase.

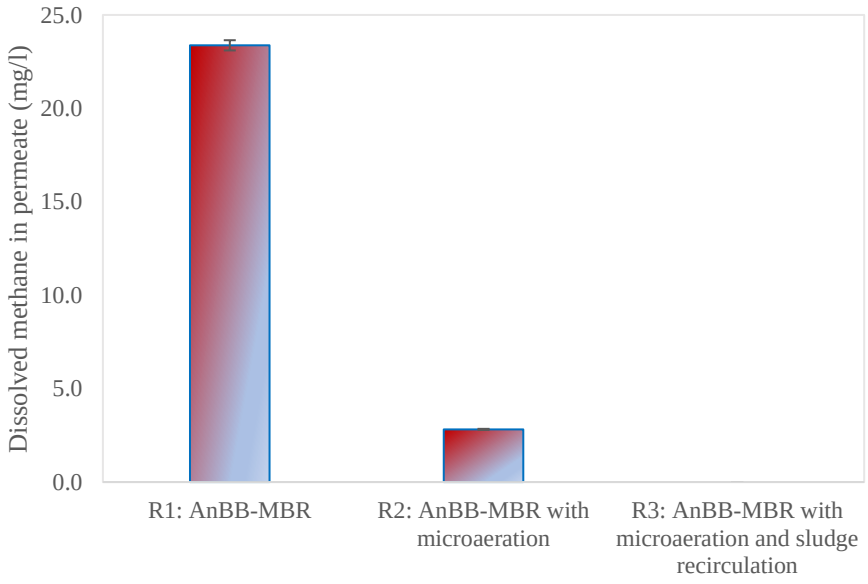


Fig. S2 Average permeated dissolved methane of all AnBB-MBRs.

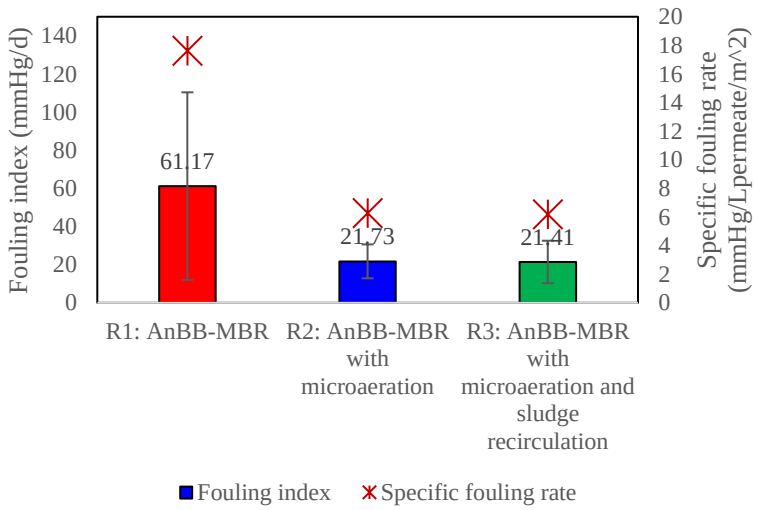


Fig. S3 Average fouling index and specific fouling rate of membrane in each AnBB-MBR

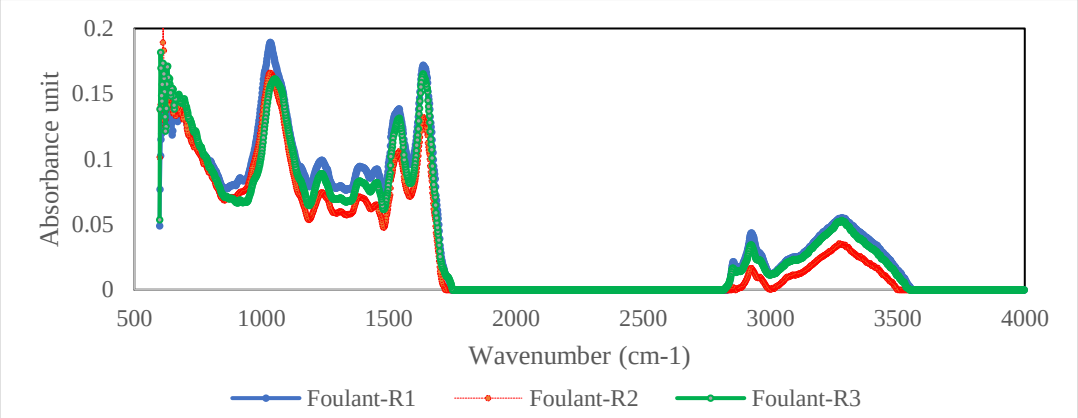


Fig. S4 FTIR of the foulants on membrane surface of each bioreactor.

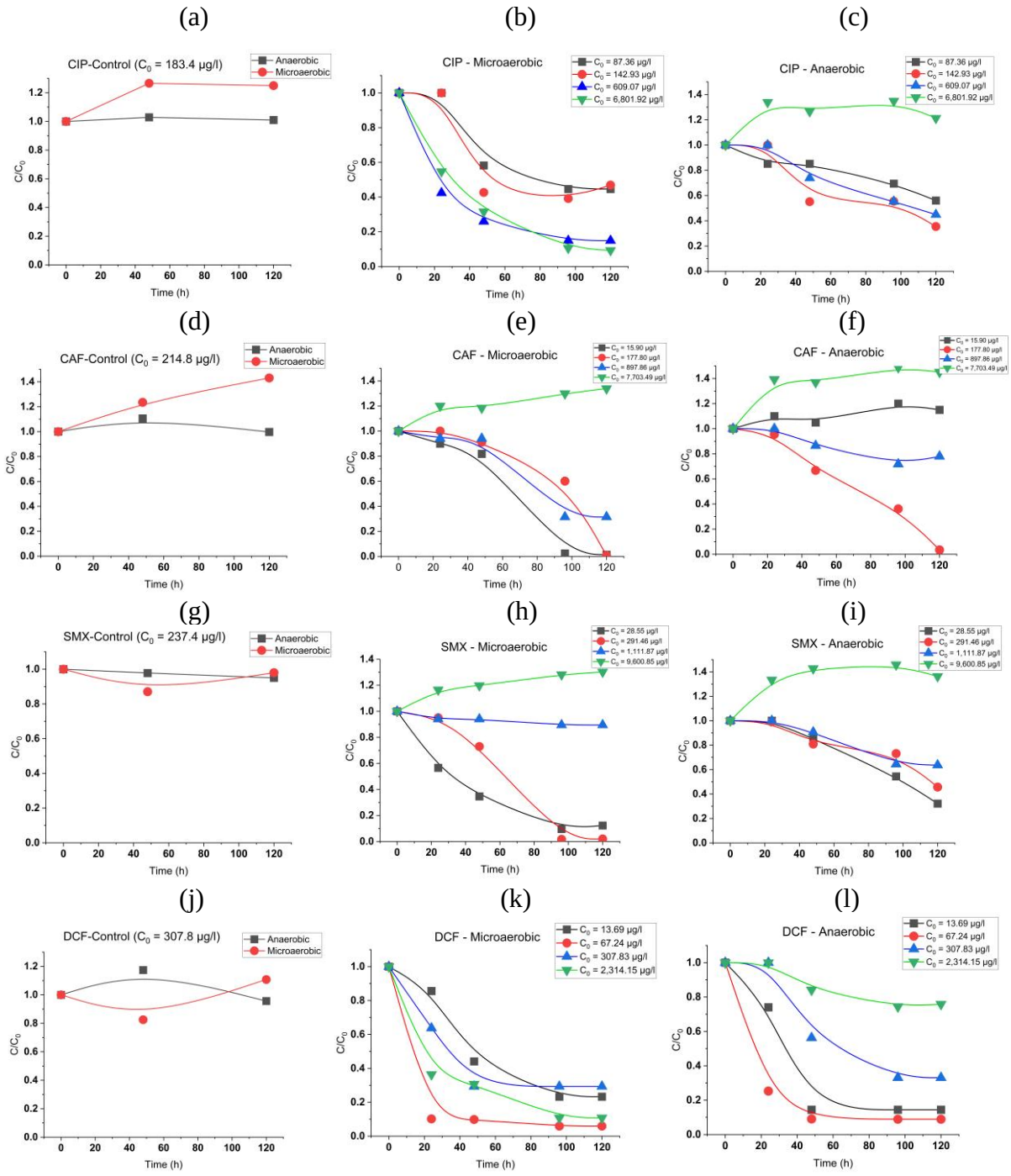
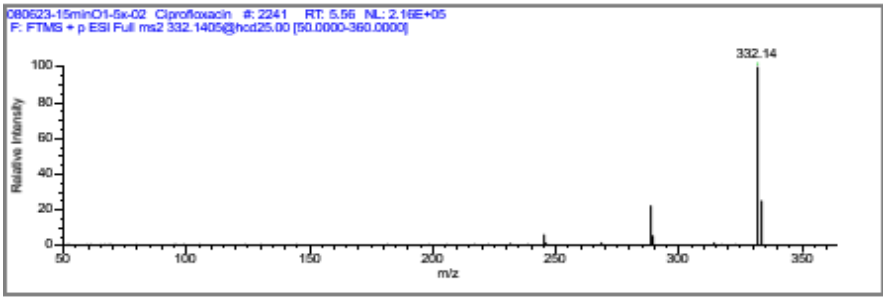


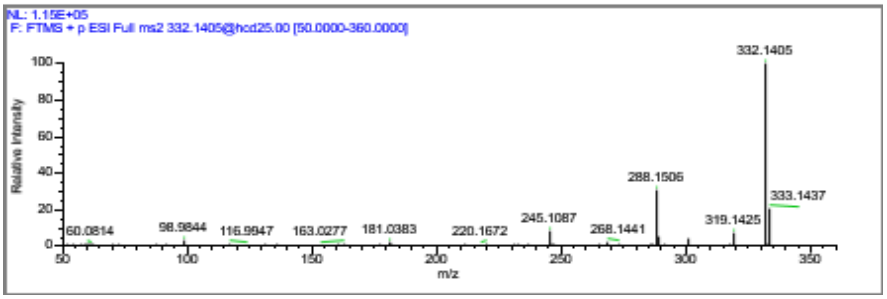
Fig. S5. Batch test for degradation kinetics of; CIP control (a), CIP microaerobic (b) and CIP anaerobic (c); CAF control (d), CAF microaerobic (e) and CAF anaerobic (f)**;** SMX control (g), SMX microaerobic (h) and SMX anaerobic (i); DCF control (j), DCF microaerobic (k) and DCF anaerobic (l).

(a)

MS-Data

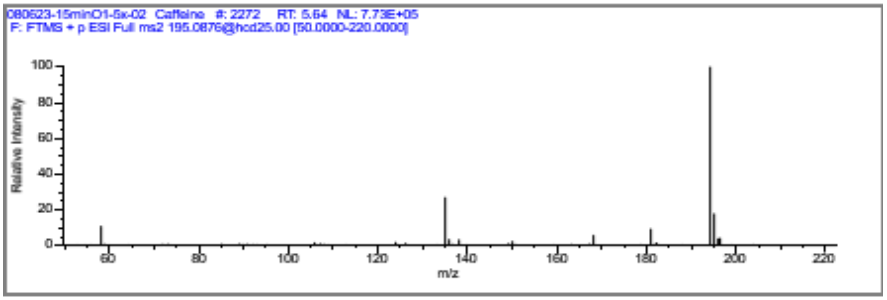


MS-Reference Std5ppb11.raw

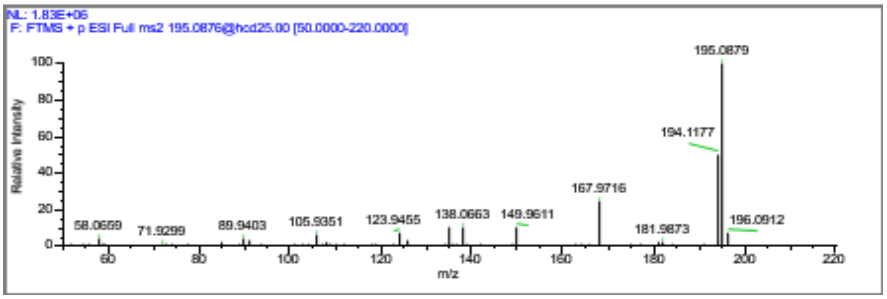


(b)

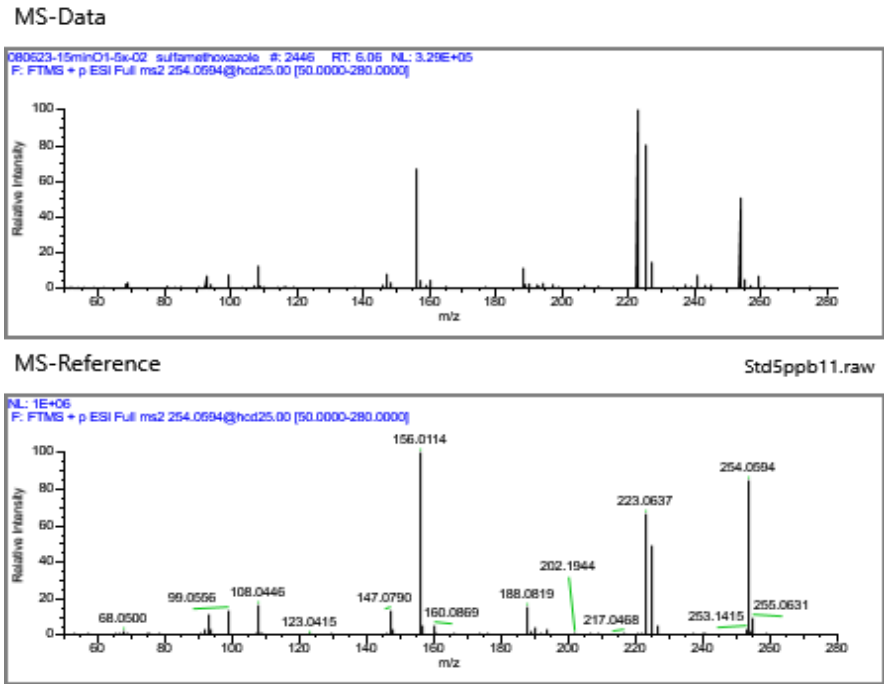
MS-Data



MS-Reference Std5ppb11.raw



(c)



(d)

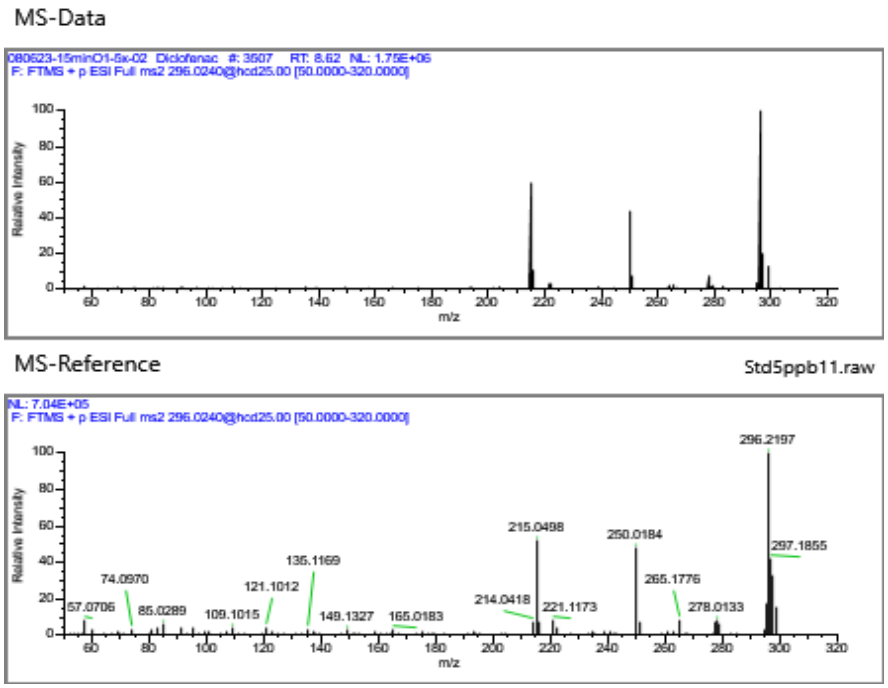


Fig. S6 MS data of targeted pharmaceuticals (a) CIP (b) CAF (c) SMX and (D) DCF.

Table S1 Profiles of measured parameters (COD, VFAs, MLSS, MLVSS, pH and ORP) observed in each position of three bioreactors and their removal efficiencies.

Parameters	Influent	R1: AnBB-MBR					R2: AnBB-MBR with microaeration					R3: AnBB-MBR with microaeration and sludge recirculation				
		C1	C2	C3	Permeate	% Removal	C1	C2	C3	Permeate	% Removal	C1	C2	C3	Permeate	% Removal
COD	245.80±94.18	140.33±5.03	101.33±8.08	n.a.	75.20±11.36	64.93±14.60	139.33±2.08	97.00±2.65	n.a.	42.90±10.48	82.50±9.80	139.83±3.49	126.33±3.06	n.a.	44.70±17.70	81.8±13.6
TKN	88.4±6.4	n.a.	n.a.	n.a.	78.0±14.8	11.11±19.93	n.a.	n.a.	n.a.	28.5±4.5	67.52±6.74	n.a.	n.a.	n.a.	27.5±17.5	67.87±14.98
NH_4^+-N	83.3±20.1	n.a.	n.a.	n.a.	75.1±10.8	9.9±31.37	n.a.	n.a.	n.a.	21.5±7.7	74.2±8.14	n.a.	n.a.	n.a.	14.3±9.7	82.9±14.73
NO_2^--N	0.01±0.01	n.a.	n.a.	n.a.	0.02±0.02	-	n.a.	n.a.	n.a.	0.36±0.42	-	n.a.	n.a.	n.a.	0.80±0.72	-
NO_3^--N	1.50±1.10	n.a.	n.a.	n.a.	2.10±1.30	-	n.a.	n.a.	n.a.	22.60±7.00	-	n.a.	n.a.	n.a.	15.20±7.70	-
PO_4^{3-}-P	11.8±6.0	n.a.	n.a.	n.a.	6.60±1.10	33.80±21.38	n.a.	n.a.	n.a.	6.70±1.40	30.64±24.64	n.a.	n.a.	n.a.	6.20±0.70	35.95±21.18
Acetic acid	34.47±0.18	2.67±0.14	<LOD	n.a.	11.16±0.11	67.62±0.49	3.67±0.22	<LOD	n.a.	n.d.	100	<LOD	<LOD	n.a.	1.06±0.34	96.9±0.26
Propionic acid	1.63±0.01	1.66±0.06	1.67±0.03	n.a.	1.71±0.08	0	1.64±0.03	1.67±0.07	n.a.	1.61±0.02	1.53±0.80	1.58±0.01	<LOD	n.a.	<LOD	100
Butyric acid	1.82±0.01	<LOD	<LOD	n.a.	1.75±0.02	4.22±1.59	<LOD	<LOD	n.a.	n.d.	100	<LOD	<LOD	n.a.	<LOD	100
Valeric acid	2.43±0.02	1.90±0.01	1.87±0.03	n.a.	2.03±0.03	16.33±0.49	1.96±0.02	1.87±0.01	n.a.	1.85±0.05	24.00±2.32	1.90±0.05	1.86±0.02	n.a.	1.84±0.02	24.20±0.02
Bottom solids (SS or MLSS)	255.83±20.61	3,120.00 ± 744.89	782.50 ± 603.51	263.33 ± 168.36	n.a.	-	1,485.00 ± 1270.24	2,455.00 ± 2274.80	1,210.00 ± 486.09	n.a.	-	4,020.00 ±3,706.64	5,770.00± 4563.96	530.00±95.29	-	-
Bottom volatile solids (VSS or MLVSS)	165.00±31.09	1,600.00 ± 190.96	547.50 ± 239.77	245.00 ± 135.76	n.a.	-	560.00 ± 167.93	450.00 ± 35.59	915.00 ± 164.53	n.a.	-	2,686.67 ±2,245.98	3,983.33± 2,421.99	503.33±116.05	-	-
VSS/SS (%)	64.31±9.20	51.28±6.56	47.51±17.78	93.04±8.08	n.a.	-	78.70±2 4.81	54.27±7.65	75.62±25.62	n.a.	-	66.83±6.25	69.04±12.51	94.97±8.08	n.a.	-
pH	7.43±0.08	7.30 ± 0.14	7.16 ± 0.05	7.28 ± 0.10	7.33 ± 0.10	-	7.45 ± 0.14	7.26 ± 0.04	6.51 ±0.15	6.61 ±0.20	-	7.39±0.14	7.31±0.07	7.06±0.23	7.13±0.12	-
ORP (mV)	n.a.	-302.2 ± 2.50	-302.0 ± 1.83	-301.5 ±1.91	n.a.	-	-300.7 ± 2.50	-301.7 ± 1.26	-89.2 ± 10.21	n.a.	-	-301.7 ±2.22	-227.2 ±2.22	-140.7 ±4.31	n.a.	-

Notes: All parameters are in unit of mg/l except pH, and n.a. is not analyzed.

Table S2 The data of LC-MS/MS Orbitrap of the targeted pharmaceuticals.

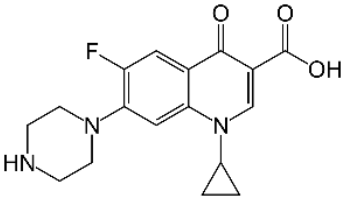
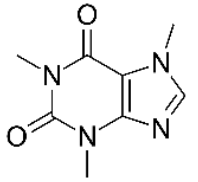
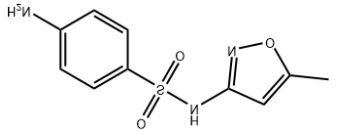
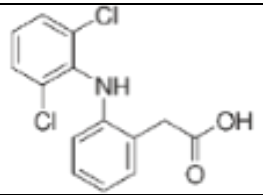
Targeted pharmaceuticals	Formular	MW (g/mol)	RT (min)	ESI	Adduct ion	m/z	m/z of fragments	LOD (ng/l)	LOQ (ng/l)	Chemical structure
Ciprofloxacin	$C_{14}H_{14}O_3$	331.34	5.58	+	M+H	332.1405	249.0663, 288.1500, 245.1082, 231.0562	44.96	149.86	
Caffeine	$C_8H_{10}N_4O_2$	194.19	5.63	+	M+H	195.0877	138.0663, 110.0716, 83.0610, 69.0455	7.13	23.75	
Sulfamethoxazole	$C_{10}H_{11}N_3O_3S$	253.28	6.06	+	M+H	254.0594	156.0117, 92.0500, 65.0393, 108.0448	2.96	9.87	
Diclofenac	$C_{14}H_{11}Cl_2O_2$	296.16	8.65	+	M+H	296.0240	215.0494, 250.0182, 278.0131	23.16	77.18	

Table S3 Sludge water distribution coefficient (Kd), logKd and zeta potential of biomass in C3 compartment of three bioreactors

Biomass position	Average zeta potential	CIP		CAF		SMX		DCF	
		Kd (L/kg.MLSS)	logKd	Kd (L/kg.MLSS)	logKd	Kd (L/kg.MLSS)	logKd	Kd (L/kg.MLSS)	logKd
C3 of R1	-22.5±0.6	505.0±256.1	2.7	503.2±366.5	2.7	70.2±28.0	1.8	55.0±17.8	1.7
C3 of R2	-19.9±0.4	1218.2±591.9	3.1	380.8±78.7	2.6	133.0±96.7	2.1	72.8±16.2	1.9
C3 of R3	-21.5±1.6	512.3±155.0	2.7	482.8±280.8	2.7	201.9±151.7	2.3	40.8±8.1	1.6

Table S4 Parameters for adsorption isotherms of targeted pharmaceuticals in batch adsorption tests.

Targeted pollutants	Tested Condition	Freundlich model			Langmuir model		
		K_f (L/g)	1/n	R^2	q_{max} (μg/g)	K_L (L/μg)	R^2
CIP	Microaerobic	6.50	0.42	0.90	101.21	0.0148	0.79
	Anaerobic	1.72	0.57	0.93	72.67	0.0079	0.88
CAF	Microaerobic	0.11	0.78	0.99	16.54	0.0049	0.98
	Anaerobic	0.24	0.69	0.99	25.41	0.0045	0.98
SMX	Microaerobic	0.01	1.03	0.99	Not fit		
	Anaerobic	0.05	0.84	0.99	27.31	0.0012	0.99
DCF	Microaerobic	0.12	0.79	0.99	16.40	0.0046	0.98
	Anaerobic	0.01	1.17	0.99	Not fit		